성균관대학교 SUNG KYUN KWAN UNIVERSITY(SKKU) | 메타바이오헬스학과

Week 11

Medical Image Generation AI

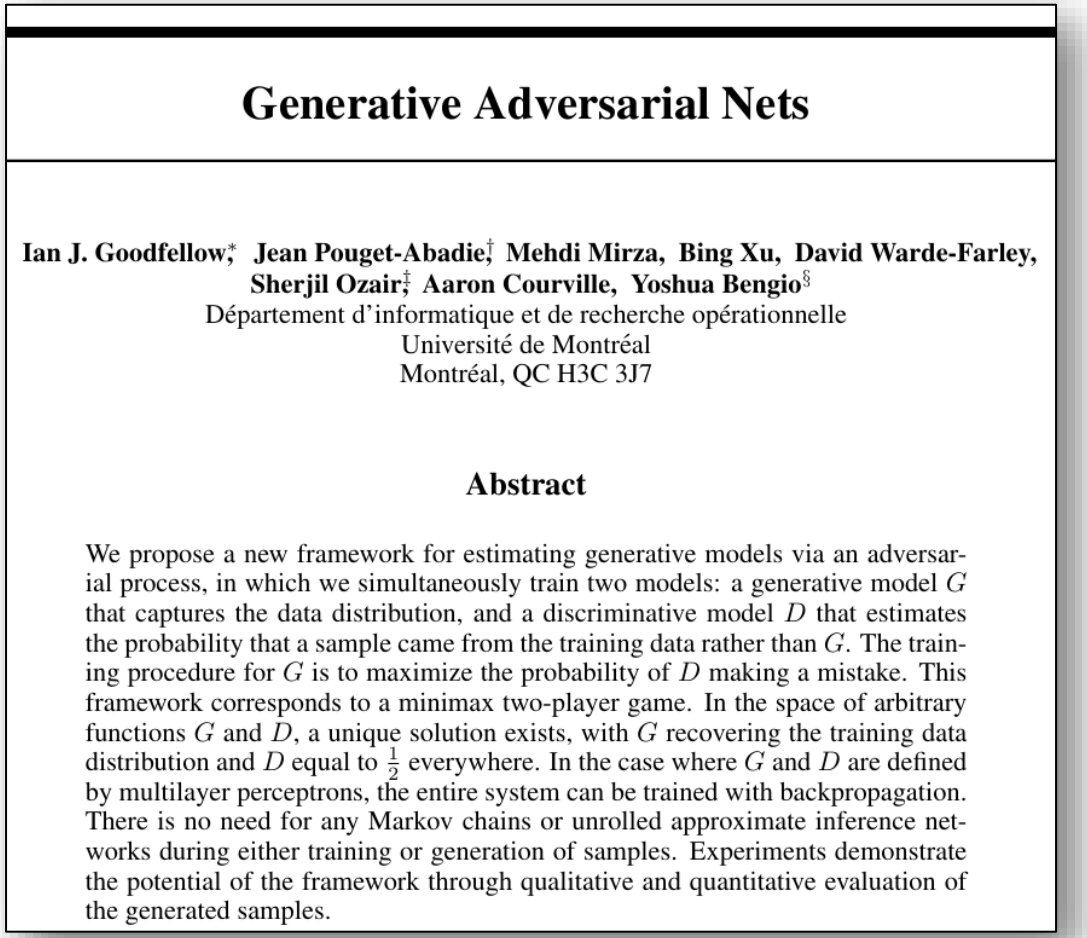
Boah Kim

Generative model

Generative adversarial network (GAN)

- Unsupervised deep learning framework
- Generate realistic synthetic images
- High accuracy in cross-modality image translation tasks

NeurIPS 2014



Generative Adversarial Nets

Ian J. Goodfellow[*], **Jean Pouget-Abadie**[†], **Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair**[‡], **Aaron Courville, Yoshua Bengio**[§]
Département d'informatique et de recherche opérationnelle
Université de Montréal
Montréal, QC H3C 3J7

Abstract

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G. The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D, a unique solution exists, with G recovering the training data distribution and D equal to $\frac{1}{2}$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

Generative model

Generative adversarial network (GAN)

- Learn patterns in input data & be utilized for generating new examples
- Useful in real-life image generation.





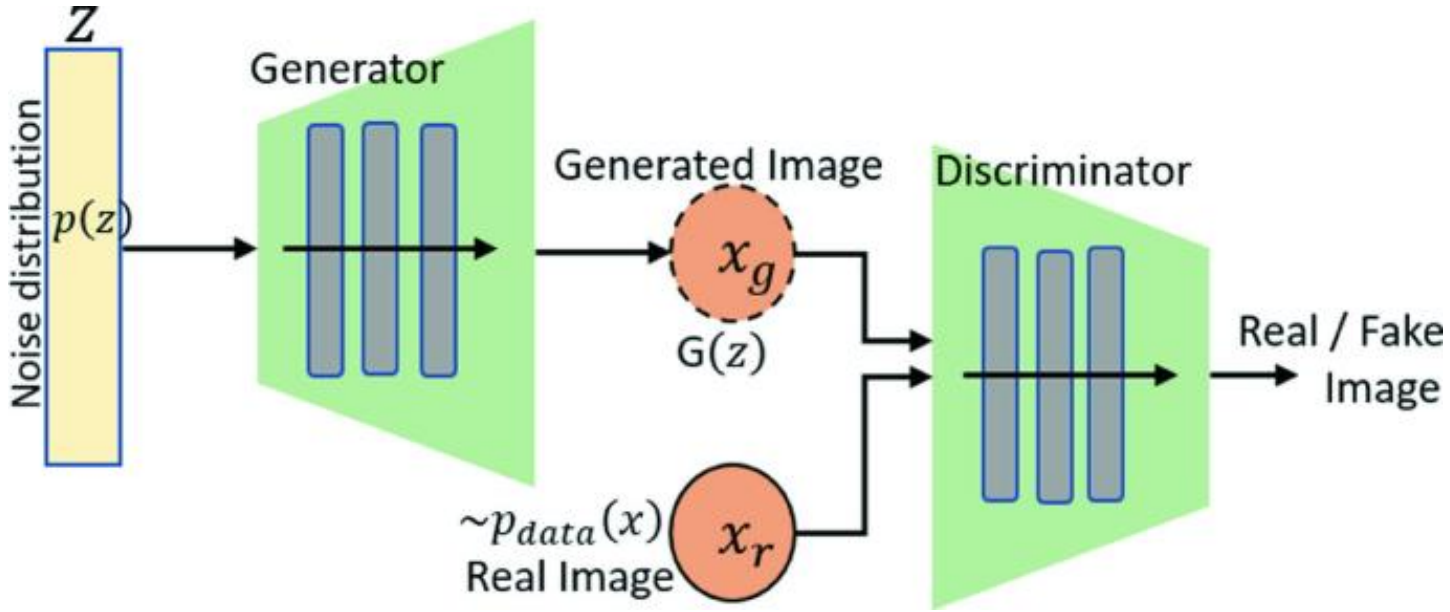
https://cs.stanford.edu/~amishkin/assets/slides/gans.pdf

Generative model

Generative adversarial network (GAN)

- Vanilla GAN
  - Generator: Takes input z from noise distribution $p(z)$ & outputs an image x_g
  - Discriminator: Takes x_g and instance of real distribution x_r

    & produces single value indicating the probability of generated sample is real and fake



https://link.springer.com/chapter/10.1007/978-981-15-9735-0_5

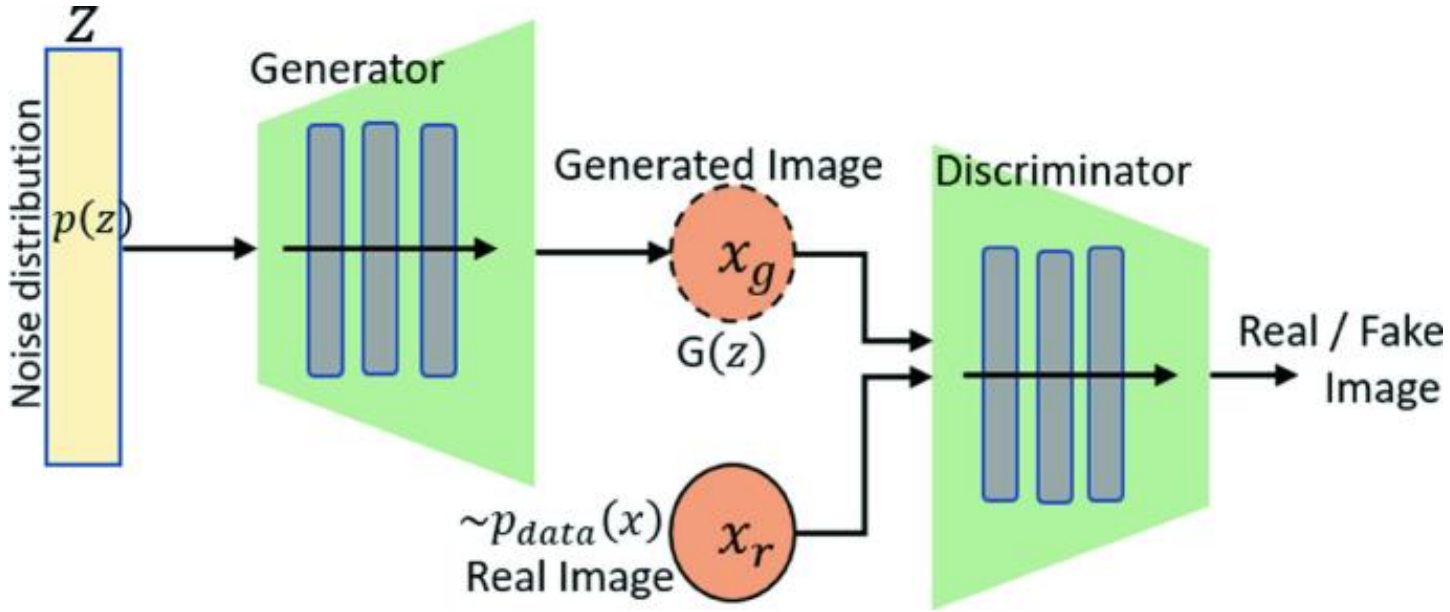
Generative model

Generative adversarial network (GAN)

- Vanilla GAN

$$\min_G \max_D V(D,G) = E_{x_r \sim p_{\text{data}(x)}} [\log D(x_r)] + E_{x_g \sim p_z(z)} [1 - \log D(G(z))]$$

  - Discriminator is to distinguish pair of real or fake samples
  - Generator is to generate a variety of realistic samples to confuse the discriminator



https://link.springer.com/chapter/10.1007/978-981-15-9735-0_5

Generative model

Generative adversarial network (GAN)

- Challenges in GANs optimization
  - Mode collapse: Generator continually produces a similar image &

    Discriminator is unable to consider the difference in generated samples

    → Generator easily fools the discriminator

    → Restricts the learning of the generator and focuses on a limited set of images

  - Vanishing gradients: When discriminator becomes too strong,

    - Discriminator does not provide enough corresponding gradients as feedback to the generator
    - Generator is unable to produce a competitive sample

    When discriminator become too weak, loss represents nothing to the generator

https://link.springer.com/chapter/10.1007/978-981-15-9735-0_5

Generative model

Generative adversarial network (GAN)

- Challenges in GANs optimization
  - Problem with counting: Early GANs framework was unable to distinguish several objects at particular location in generated image
    (more/less number of nose holes, eye's/ear's fails to place the right location)

  - Problem with perspective: Unable to understand the perspective and failed to differentiate front view and back view objects
    → Not work well at the conversion of 3D into 2D representation

https://link.springer.com/chapter/10.1007/978-981-15-9735-0_5

Generative model

Image-to-Image Translation with Conditional Adversarial Networks (Pix2Pix) CVPR 2017

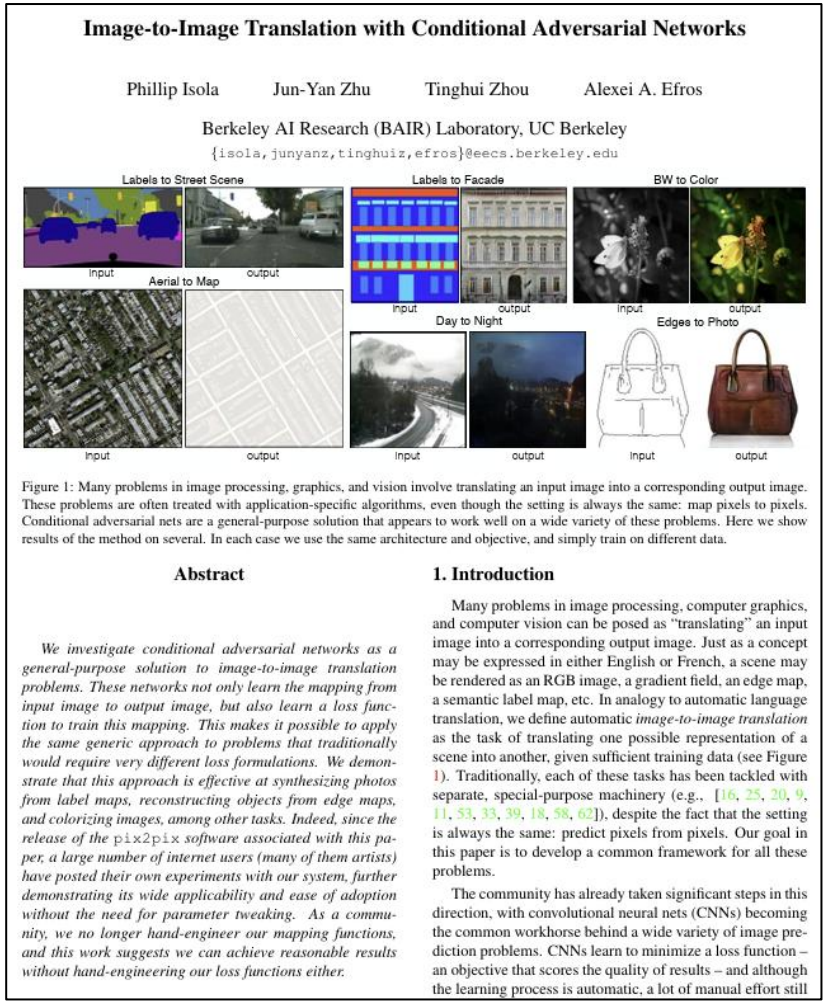


Image-to-Image Translation with Conditional Adversarial Networks

Phillip Isola Jun-Yan Zhu Tinghui Zhou Alexei A. Efros

Berkeley AI Research (BAIR) Laboratory, UC Berkeley

{isola,junyanz,tinghuiz,efros}@eecs.berkeley.edu

Figure 1: Many problems in image processing, graphics, and vision involve translating an input image into a corresponding output image. These problems are often treated with application-specific algorithms, even though the setting is always the same: map pixels to pixels. Conditional adversarial nets are a general-purpose solution that appears to work well on a wide variety of these problems. Here we show results of the method on several. In each case we use the same architecture and objective, and simply train on different data.

Abstract

We investigate conditional adversarial networks as a general-purpose solution to image-to-image translation problems. These networks not only learn the mapping from input image to output image, but also learn a loss function to train this mapping. This makes it possible to apply the same generic approach to problems that traditionally would require very different loss formulations. We demonstrate that this approach is effective at synthesizing photos from label maps, reconstructing objects from edge maps, and colorizing images, among other tasks. Indeed, since the release of the `pix2pix` *software associated with this paper, a large number of internet users (many of them artists) have posted their own experiments with our system, further demonstrating its wide applicability and ease of adoption without the need for parameter tweaking. As a community, we no longer hand-engineer our mapping functions, and this work suggests we can achieve reasonable results without hand-engineering our loss functions either.*

1. Introduction

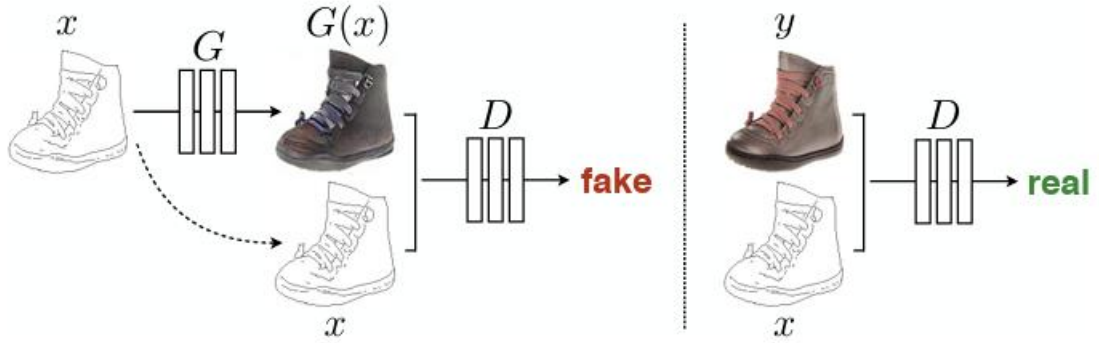
Many problems in image processing, computer graphics, and computer vision can be posed as "translating" an input image into a corresponding output image. Just as a concept may be expressed in either English or French, a scene may be rendered as an RGB image, a gradient field, an edge map, a semantic label map, etc. In analogy to automatic language translation, we define automatic *image-to-image translation* as the task of translating one possible representation of a scene into another, given sufficient training data (see Figure 1). Traditionally, each of these tasks has been tackled with separate, special-purpose machinery (e.g., [16, 25, 20, 9, 11, 53, 33, 39, 18, 58, 62]), despite the fact that the setting is always the same: predict pixels from pixels. Our goal in this paper is to develop a common framework for all these problems.

The community has already taken significant steps in this direction, with convolutional neural nets (CNNs) becoming the common workhorse behind a wide variety of image prediction problems. CNNs learn to minimize a loss function – an objective that scores the quality of results – and although the learning process is automatic, a lot of manual effort still

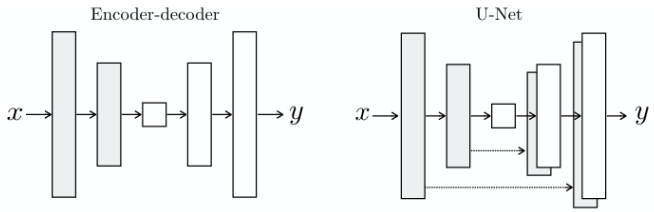
- A supervised conditional GAN framework
- Learns mappings from input images to output images
- Suitable for a variety of paired image translation tasks.

Generative model

Image-to-Image Translation with Conditional Adversarial Networks (Pix2Pix) CVPR 2017



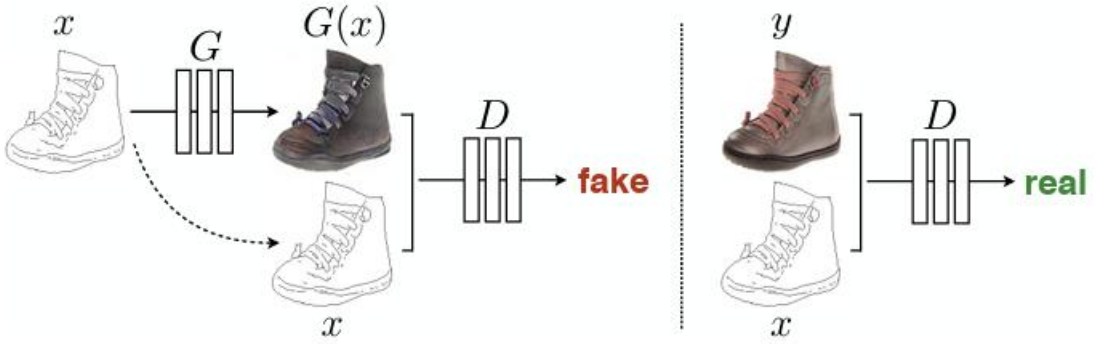
- Generator (G)
  - Encodes the input image and decodes it into a target image.
  - Uses skip connections to retain spatial details.



- Discriminator (PatchGAN):
  - Evaluates local image patches instead of the entire image.
  - Determines whether each patch is real or generated.

Generative model

Image-to-Image Translation with Conditional Adversarial Networks (Pix2Pix) CVPR 2017



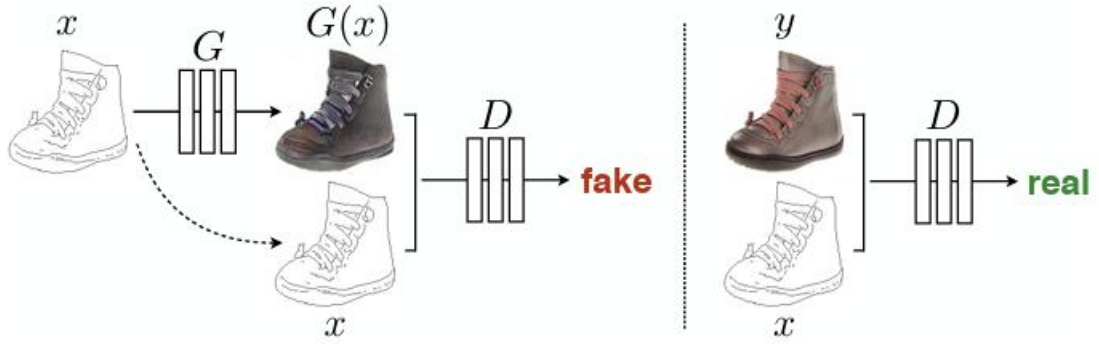
- Training procedure
  - Requires paired training data: (input image → target image).
  - Jointly optimizes adversarial loss and pixel-wise L1 loss.

$$G^*, D^* = \arg\min_G \max_D L_{\text{CGAN}}(G, D) + \lambda L_{L1}(G)$$

$$L_{\text{CGAN}}(G, D) = E_{x,y}[\log D(x, y)] + E_{x,z}[\log(1 - D(x, G(x, z)))] \qquad L_{L1}(G) = E_{x,y\sim P\text{data}(x,y),z\sim P(z)}[||y - G(x, z)||_1]$$

Generative model

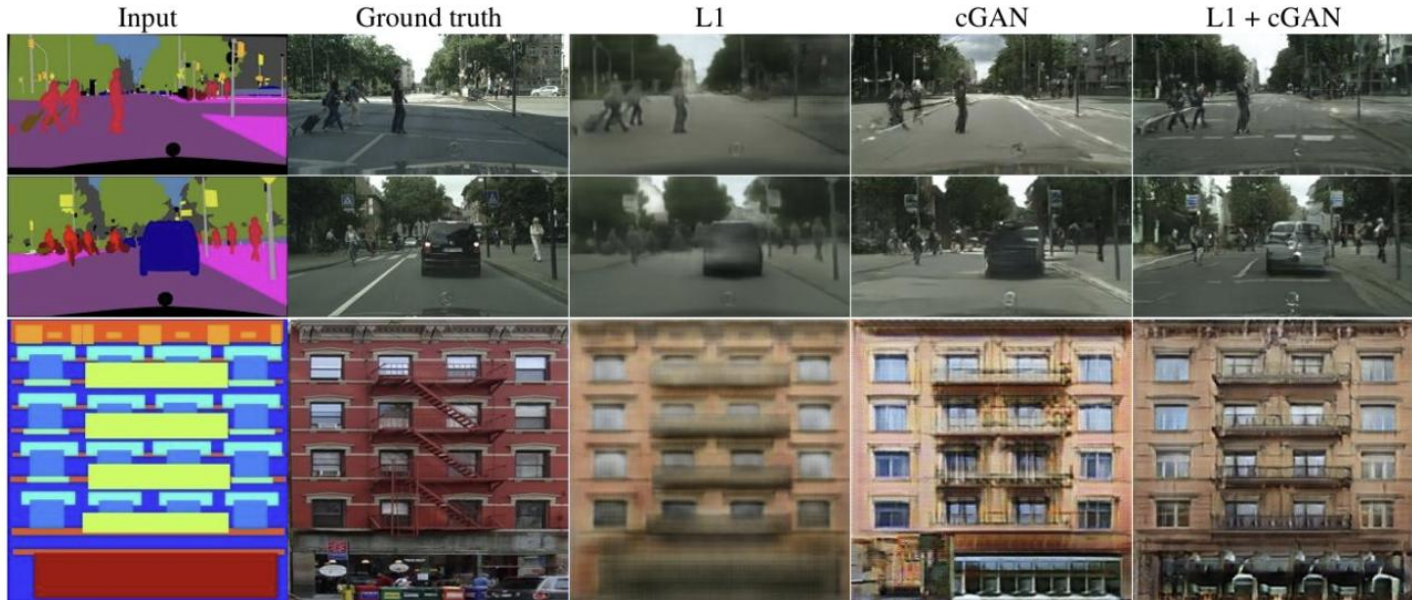
Image-to-Image Translation with Conditional Adversarial Networks (Pix2Pix) CVPR 2017



- Training procedure
  - Requires paired training data: (input
  - Jointly optimizes adversarial loss and

$$G^*, D^* = \text{a}$$

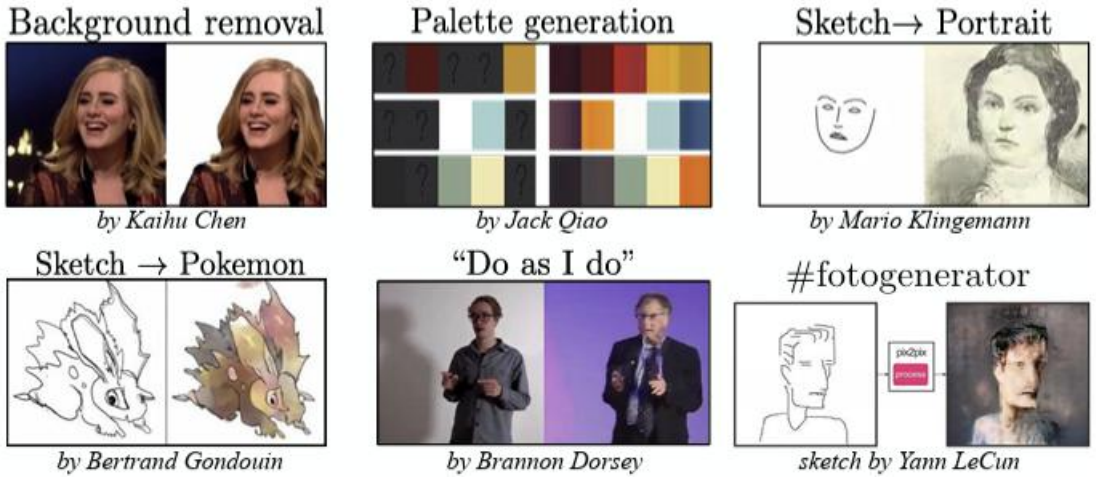
$$L_{\text{CGAN}}(G, D) = E_{x,y}[\log D(x, y)] + E_{x,z}[\text{l}$$



Generative model

Image-to-Image Translation with Conditional Adversarial Networks (Pix2Pix) CVPR 2017

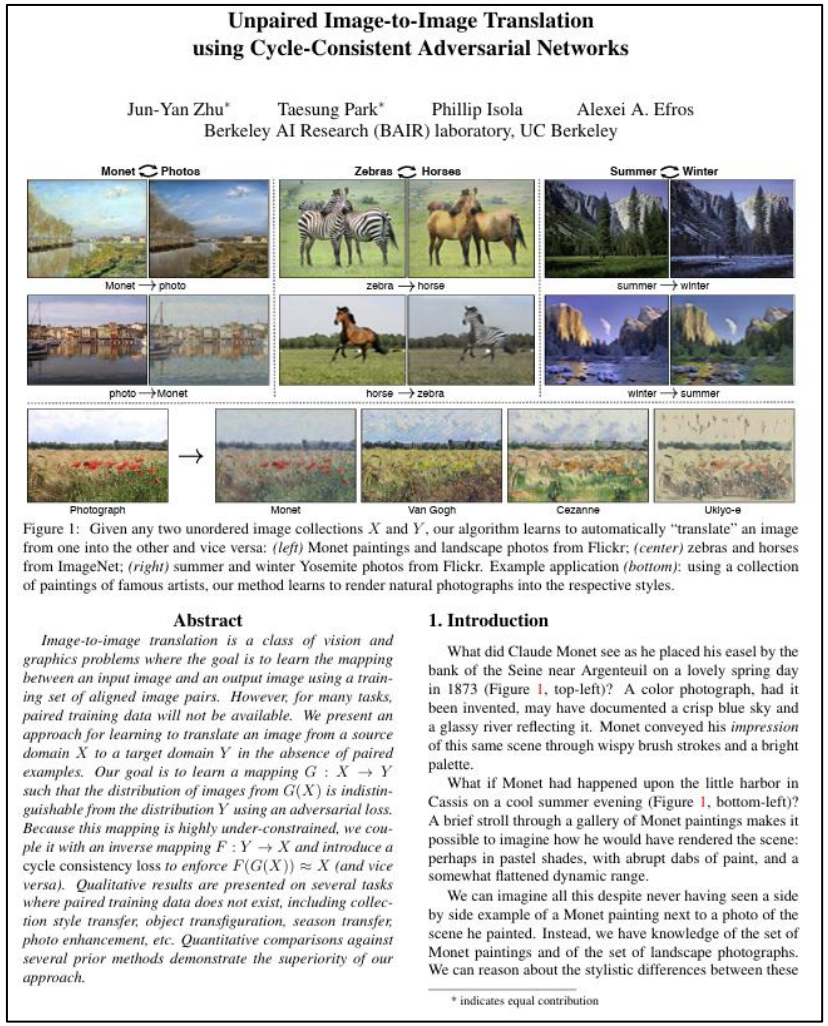
- Advantages
  - Supervised Framework: Clearly defined target outputs lead to stable and precise learning.
  - Local patch evaluation (PatchGAN) enhances image detail and sharpness.
  - Applicable to diverse tasks across domains without significant model adjustments.





Generative model

Cycle-Consistent Generative Adversarial Networks (CycleGAN) CVPR 2017



Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

Jun-Yan Zhu* Taesung Park* Phillip Isola Alexei A. Efros
Berkeley AI Research (BAIR) laboratory, UC Berkeley

Figure 1: Given any two unordered image collections X and Y, our algorithm learns to automatically "translate" an image from one into the other and vice versa: *(left)* Monet paintings and landscape photos from Flickr; *(center)* zebras and horses from ImageNet; *(right)* summer and winter Yosemite photos from Flickr. Example application *(bottom)*: using a collection of paintings of famous artists, our method learns to render natural photographs into the respective styles.

Abstract

Image-to-image translation is a class of vision and graphics problems where the goal is to learn the mapping between an input image and an output image using a training set of aligned image pairs. However, for many tasks, paired training data will not be available. We present an approach for learning to translate an image from a source domain X to a target domain Y in the absence of paired examples. Our goal is to learn a mapping $G : X \rightarrow Y$ such that the distribution of images from $G(X)$ is indistinguishable from the distribution Y using an adversarial loss. Because this mapping is highly under-constrained, we couple it with an inverse mapping $F : Y \rightarrow X$ and introduce a cycle consistency loss *to enforce $F(G(X)) \approx X$ (and vice versa). Qualitative results are presented on several tasks where paired training data does not exist, including collection style transfer, object transfiguration, season transfer, photo enhancement, etc. Quantitative comparisons against several prior methods demonstrate the superiority of our approach.*

1. Introduction

What did Claude Monet see as he placed his easel by the bank of the Seine near Argenteuil on a lovely spring day in 1873 (Figure 1, top-left)? A color photograph, had it been invented, may have documented a crisp blue sky and a glassy river reflecting it. Monet conveyed his *impression* of this same scene through wispy brush strokes and a bright palette.

What if Monet had happened upon the little harbor in Cassis on a cool summer evening (Figure 1, bottom-left)? A brief stroll through a gallery of Monet paintings makes it possible to imagine how he would have rendered the scene: perhaps in pastel shades, with abrupt dabs of paint, and a somewhat flattened dynamic range.

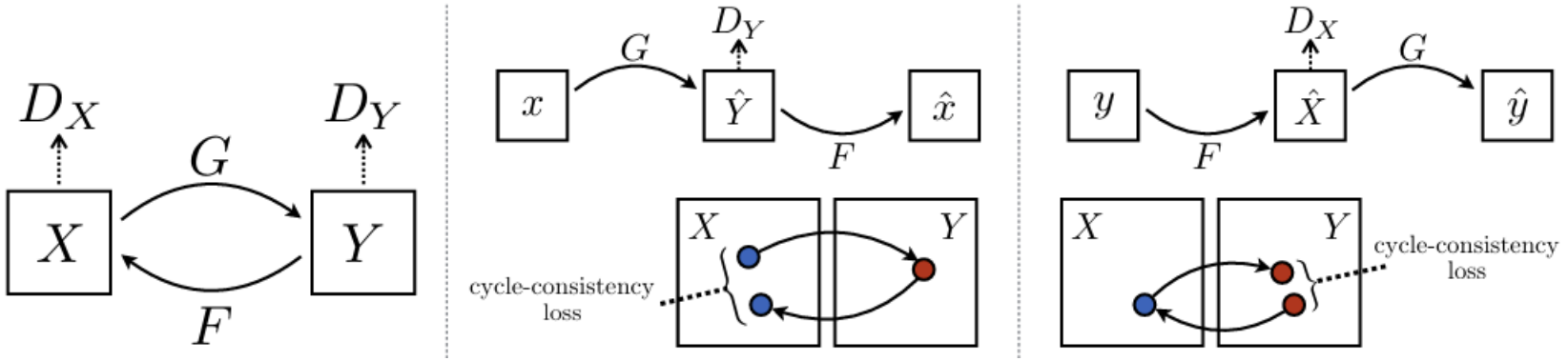
We can imagine all this despite never having seen a side by side example of a Monet painting next to a photo of the scene he painted. Instead, we have knowledge of the set of Monet paintings and of the set of landscape photographs. We can reason about the stylistic differences between these

* indicates equal contribution

- Unsupervised GAN framework
- Learns to translate images from one domain to another without paired data.
- Widely adopted for diverse image translation tasks.

Generative model

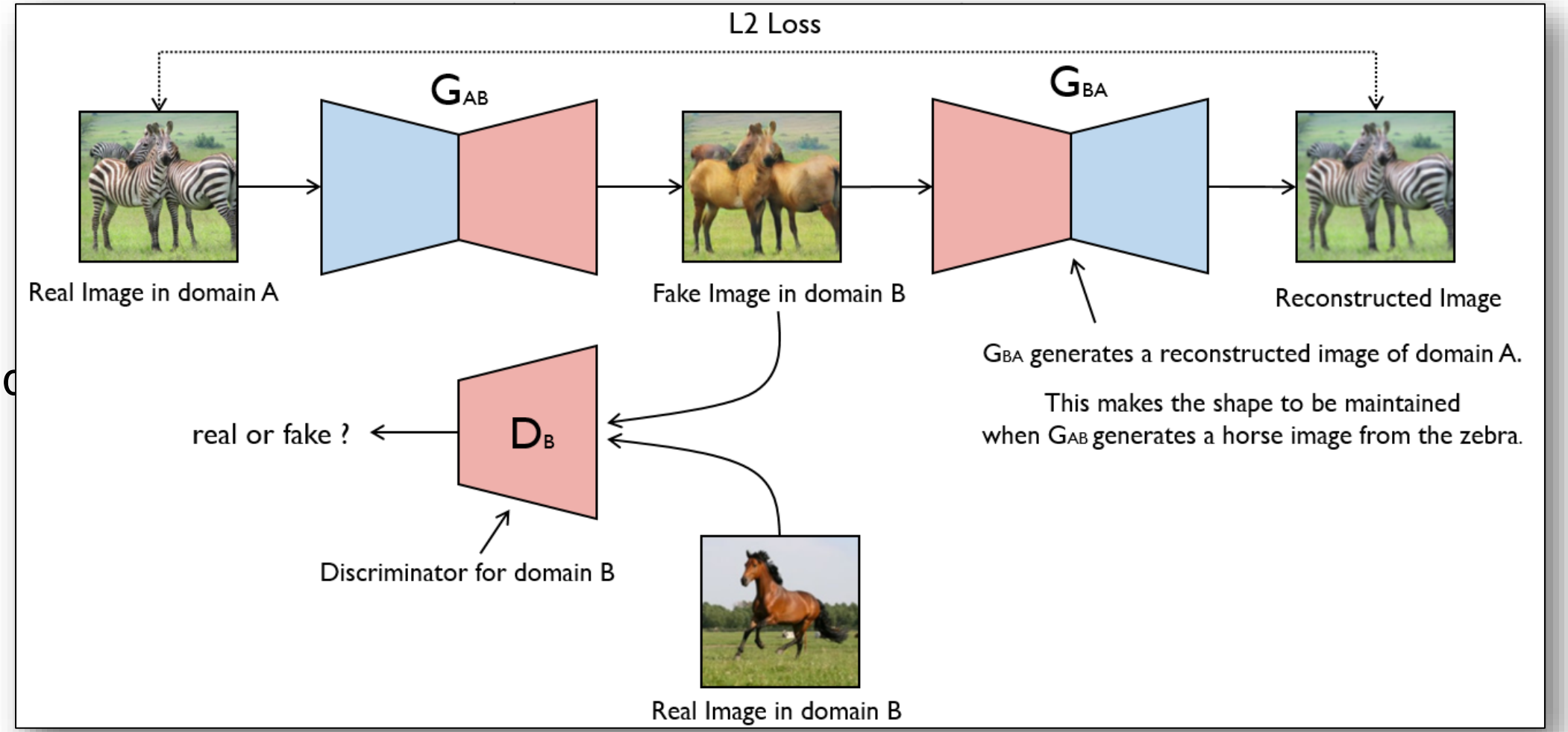
Cycle-Consistent Generative Adversarial Networks (CycleGAN) CVPR 2017



- Consists of two generators and two discriminators
  - Generator $G{:}X \rightarrow Y$: Generates an image in Y domain from X domain
  - Generator $F{:}Y \rightarrow X$: Generates an image in X domain from Y domain
  - Discriminator D_X : Distinguishes between real X and synthetic X
  - Discriminator D_Y : Distinguishes between real Y and synthetic Y

Generative model

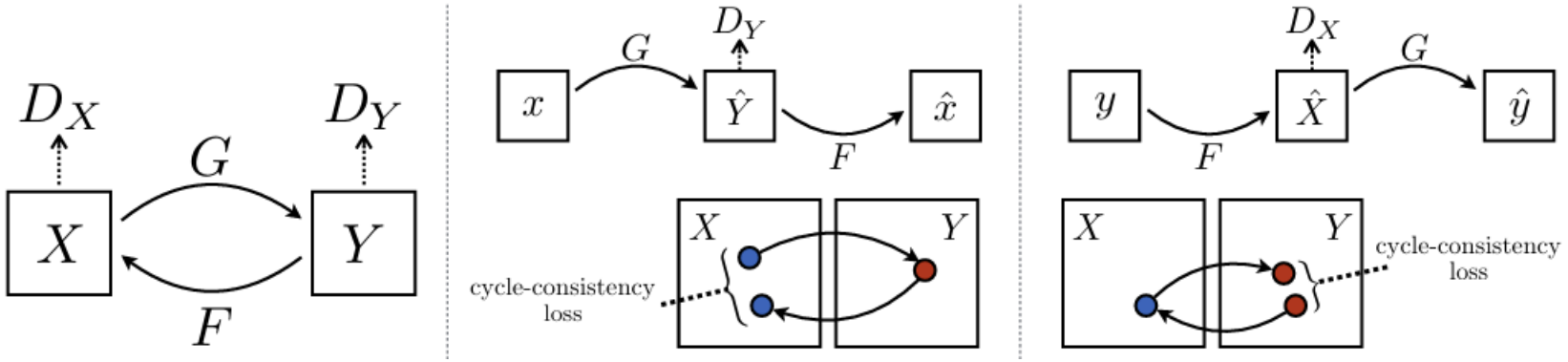
Cycle-Consistent Generative Adversarial Networks (CycleGAN) CVPR 2017



- C

Generative model

Cycle-Consistent Generative Adversarial Networks (CycleGAN) CVPR 2017



- Cycle-consistency loss
  - Ensures that translated images can be reconstructed back to original domain.

$$\mathcal{L}(G, F, D_X, D_Y) = \mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) + \mathcal{L}_{\text{GAN}}(F, D_X, Y, X) + \lambda \mathcal{L}_{\text{cyc}}(G, F),$$

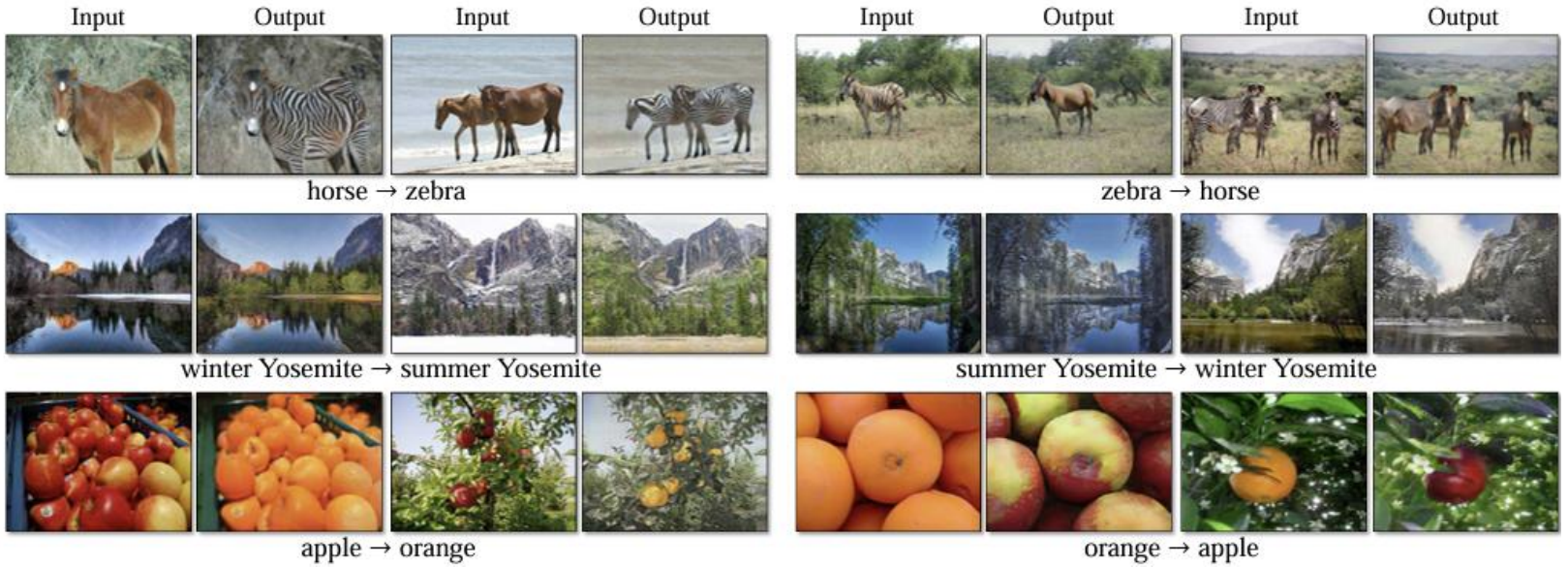
$$\mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) = \mathbb{E}_{y \sim p_{\text{data}}(y)}[\log D_Y(y)] + \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log(1 - D_Y(G(x))]$$

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{\text{data}}(y)}[\|G(F(y)) - y\|_1].$$

Generative model

Cycle-Consistent Generative Adversarial Networks (CycleGAN) CVPR 2017

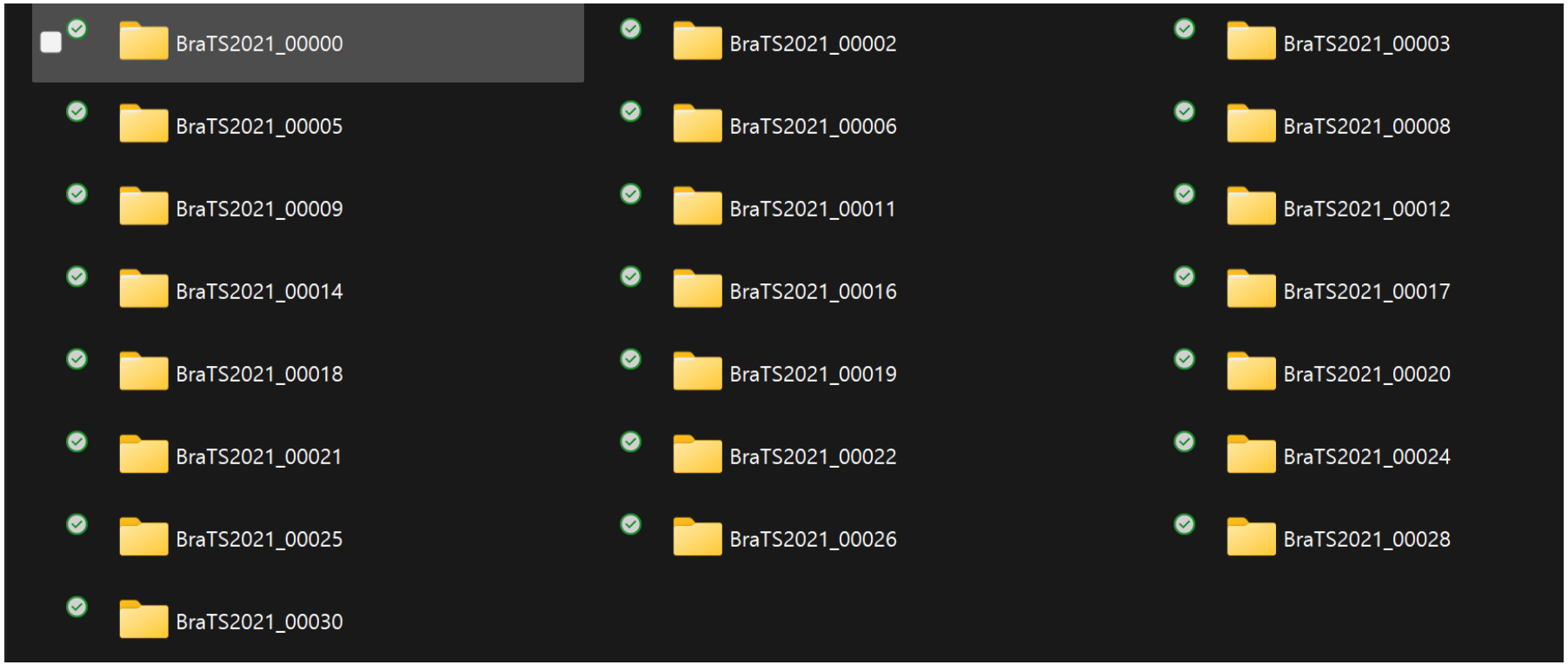
- Advantages
  - Unsupervised Training (Does not require paired images)
  - Easier and cheaper to collect training data.
  - Works well across significantly different image domains.
  - Maintains structural integrity and preserves content features during translation.



Project Assignment

Multi-parametric brain MRI generation

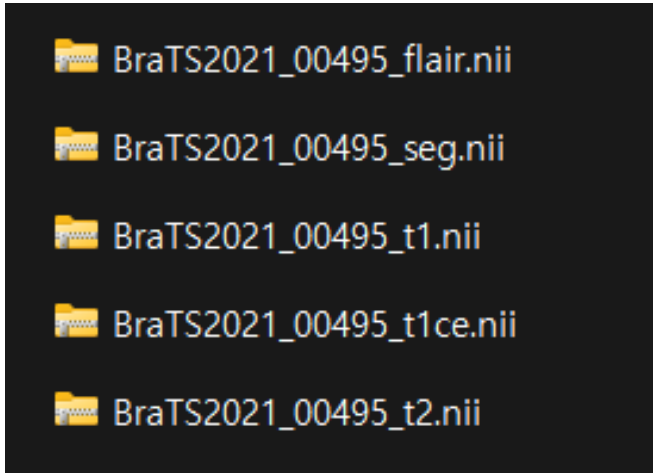
- To translate T1-weighted to T2-weighted, or T2-weighted to T1-weighted
- Dataset: Brain MRI (BRaTS 2021 Task 1 dataset)
  - Download the dataset from I-Campus:



Project Assignment

Multi-parametric brain MRI generation

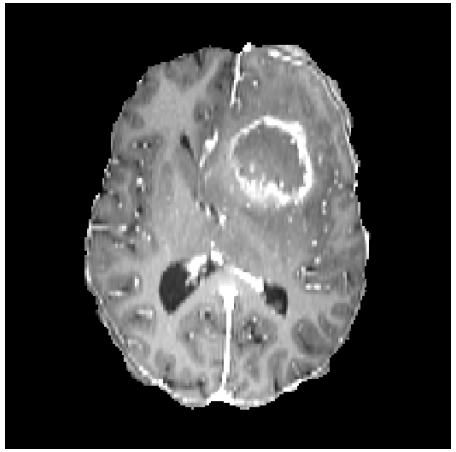
- To translate T1-weighted to T2-weighted, or T2-weighted to T1-weighted
- Dataset: Brain MRI (BRaTS 2021 Task 1 dataset)
  - All BraTS mpMRI scans are available as NIfTI files (.nii.gz)
    - a) native (T1)
    - b) **post-contrast T1-weighted (T1ce),**
    - **c) T2-weighted (T2),**
    - d) T2 Fluid Attenuated Inversion Recovery (FLAIR)



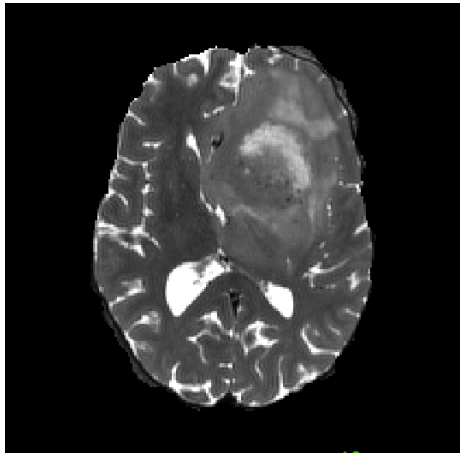
Project Assignment

Multi-parametric brain MRI generation

- Data setting
  - Only work with ***_T1ce.nii.gz and *_t2.nii.gz** for each subject.
  - In our project, we are going to use the following 4 data for test set: 00025, 00026, 00028, and 00030.
    - DO NOT use these sets for training.
  - In training, shuffle images for the training data to be unpaired.



post-contrast T1-weighted

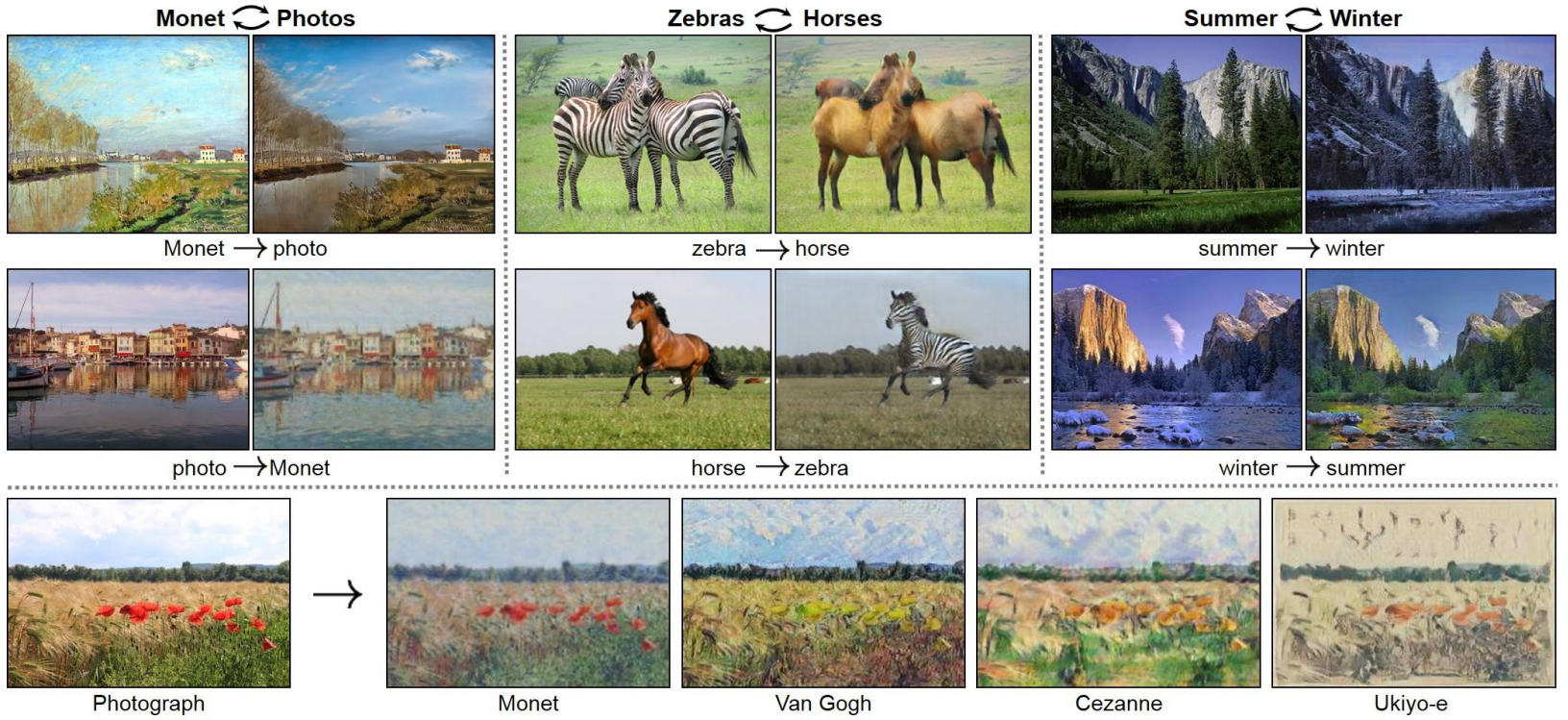


T2-weighted

Project Assignment

Multi-parametric brain MRI generation

- Network
  - Use CycleGAN model using PyTorch
  - You may also check the following link for reference: https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix



Project Assignment

Multi-parametric brain MRI generation

- Students will be required to:
  - Handle the data (preprocess, upload to Colab , separate between train, validation, test sets, etc)
  - Implement training codes
    - Use default settings used in the source code, except for generators.
  - Evaluate the performance of the networks: implement PSNR and SSIM metrics for evaluation
  - Report on the methods and results

Project Assignment

Multi-parametric brain MRI generation

- Captured images of your code (network, training scheme, metrics implementation, etc)
- Brief explanation of your code
- Compare two different generators: resnet_6blocks & unet_128.
- Provide 3 visual examples of your outputs
  - Each example includes [Real T1w, Real T2w, Synthetic T2w, Synthetic T1w].
- Report your results in quantitative metrics. PSNR and SSIM metric is required. Extra measurements for further analysis are welcome.

Project Assignment

Multi-parametric brain MRI generation

- **Format:**
  - Does not have to be formal. As long as your report has the aforementioned points, it will be fine.
  - **ONLY PDF** documents. Please name the file as: StudentID_Name_Report.pdf
  - Due date: 2026.11.22 23:59

- **You will not be graded on the performance of the network.** Your implementation process, analysis on source of errors, and varying technique attempts are more important.

Q & A